The Ag-Si (Silver-Silicon) System

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Equilibrium Diagram

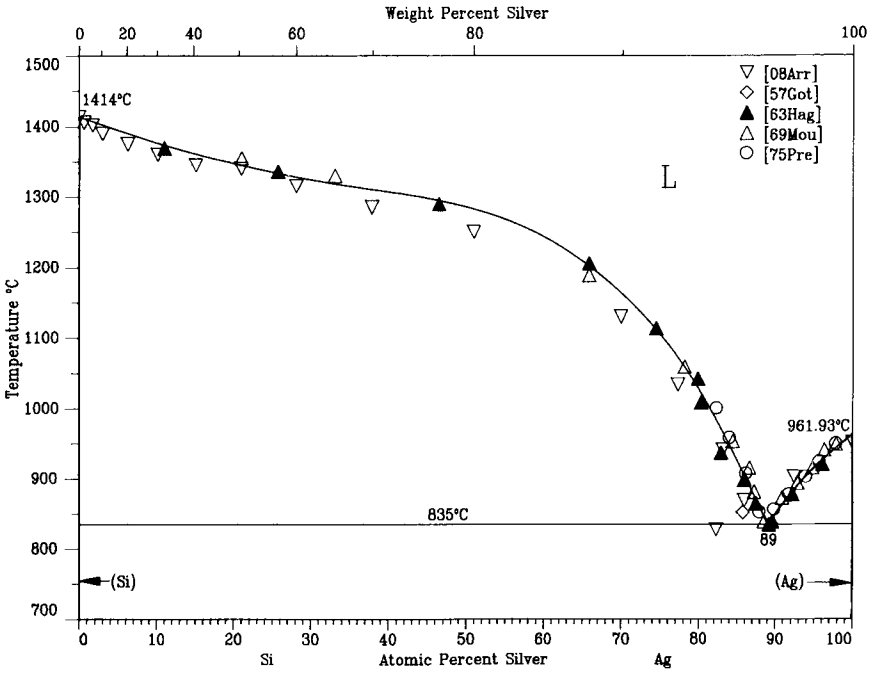
The assessed equilibrium phase diagram of the Si-Ag system is presented in Fig. 1. The system has a eutectic reaction around 845 °C and 89 at.% Ag. The mutual solubilities of Si and Ag in the solid state are negligible.

The main features of the Si-Ag phase diagram were determined first by [08Arr] by means of thermal analysis. The measurements were of understandably low accuracy: the melting temperature of Ag was reported 10 °C below the now accepted value (961.93 °C [Melt]), and the eutectic point differs considerably from coordinates determined more recently. The most comprehensive thermal analysis study of the system was performed by [63Hag], who heated Si-Ag alloys to about 100 to 200 °C above the estimated liquidus temperature, cooled them at an average rate of 2 to 7 °C/min, and determined the liquidus and eutectic temperatures from the cooling curves; the measurements were conducted using 99.86% pure Si and 99.95% pure Ag in an alumina crucible under an atmosphere of Ar. The liquidus coordinates, which were claimed to be accurate within 1 °C and 0.3 at.%, are reproduced in Tables 1 and 2 and Fig. 1; the temperatures are averages of two reported measurements for each composition. Data of [63Hag] are in good agreement with thermal analysis results obtained by other investigators [69Mou, 75Pre]. [69Mou] took their cooling curves at a rate of 5 °C/min and used 99.9% pure Si with Ag of "spectrographic purity;" [75Pre] conducted their measurements only in the Ag-rich side of the diagram, using 99.995% Ag and Si of unspecified purity. Results of both references are included in Tables 1 and 2 and Fig. 1, those of [75Pre] having been transcribed from graphical data.

The coordinates of the eutectic point reported by various investigators are listed in Table 3, together with those assessed on the basis of experimental data and the thermodynamic calculations presented later.

The (Si) solidus in the high-temperature range is reproduced in Fig. 2 [61Bol]. The authors determined solid solubilities from radiotracer diffusion experiments performed in quartz ampules in an atmosphere of He. The solidus is retrograde in character with a maximum solubility of 4×10^{-4} at.% Ag at about 1350

Fig. 1 Assessed Si-Ag Phase Diagram



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°C. The reported data agree with the earlier assessment of [57Col], indicating the solubility to be in the range 10^{-6} to 6×10^{-5} at.% Ag at 1200 °C.

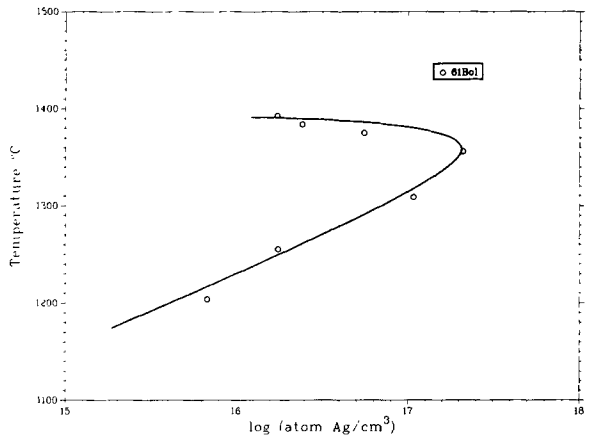
Metastable Phases

The occurrence of two metastable phases was reported in the Ag-rich side of the diagram: a cph structure in the composition range 75 to 95 at.% Ag [65Luo, 66Ana]; and an orthorhombic phase (Ag_2Si) containing about 70 at.% Ag [74Sur]. Both phases were obtained in alloys solidified rapidly at cooling rates above 10^5 °C/s by the melt spinning technique on Cu substrates. The cph phase starts decomposing above 100 °C [65Lou], and the orthorhombic phase decomposes to the equilibrium phases upon heating to "high temperatures" [74Sur].

The lattice parameters of the metastable phases are presented in Table 4. The structure of the orthorhombic phase described in [74Sur] was also indexed by him on the basis of the tetragonal unit cell (a = 0.5556 and c = 0.8340 nm), but this indexing did not prove satisfactory for all the interplanar spacings observed. No evidence of extended solid solubility of Si in (Ag) was discovered [65Luo], although a decrease of Ag lattice parameter to 0.40795 nm in the rapidly quenched alloys was interpreted by [74Sur] as implying that a small amount of Si does dissolve in (Ag).

The crystallization of Si-Ag amorphous films vapor deposited on NaCl substrates was investigated by [72Her]. The films were prepared in the form of a triple layer sandwich of 40 nm Si between two 25 nm films of Ag. The crystallization temperature of Si was reported as 540 °C.

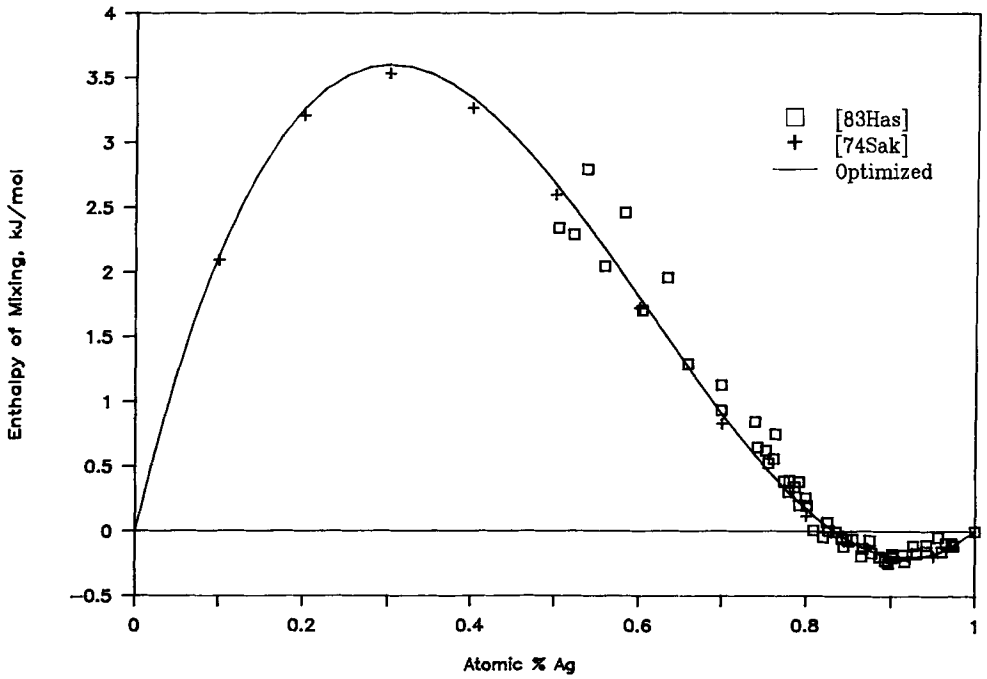
Fig. 2 Solid Solubility of Ag in (Si)



From [61Bol].

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Fig. 3 Si-Ag Liquid Enthalpy of Mixing



From [83Has] and [74Sak].

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